

Mihaela Cătălina Stoian

Department of Electrical and Electronic Engineering – Imperial College London

✉ m.stoian@imperial.ac.uk 🏠 mihaela-stoian.github.io 🌐 LinkedIn 🎓 Google Scholar

Research Interests

I build neuro-symbolic AI systems that guarantee safety and compliance by injecting strict domain requirements into neural networks. My focus areas include tabular data synthesis, autonomous driving, and constraining LLMs for theorem proving.

Neuro-Symbolic AI • Knowledge-Aided Machine Learning • Deep Generative Modelling • Large Language Models

Education

University of Oxford

DPhil in Computer Science

Oxford, UK

2021–2026

- **Thesis:** Knowledge-Refined Deep Neural Networks for Real-World Applications
- **Supervisor:** Prof. Thomas Lukasiewicz
- Developed the **first frameworks** that make deep generative models for tabular data aligned and provably compliant with background knowledge, expressed as linear constraints and disjunctions of linear inequalities capturing non-convex and even disconnected feasible spaces.
- **Areas:** deep learning, generative modelling, autonomous driving, neuro-symbolic AI.

The University of Edinburgh

MInf (Master of Informatics), First Class Honours

Edinburgh, UK

2014–2019

- **Master's thesis:** Improving extrinsic performance in zero-resource settings
- **Bachelor's thesis:** Multilingual bottleneck features for zero-resource speech processing
- **Supervisor:** Prof. Sharon Goldwater
- **Areas:** NLP, speech-to-text machine translation, speech processing.

Selected Publications

- [1] J. O. J. Leang, Y. Zhao, **M. C. Stoian**, Q. Xu, H. Li, W. Li, S. B. Cohen, E. Giunchiglia. Pythagoras-Prover: Advancing Efficient Formal Proving via Augmented Lean Formalisation, arXiv 2606.12594 Technical Report, 2026. [\[Link\]](#)
- [2] J. O. J. Leang, Y. Zhao, **M. C. Stoian**, W. Li, S. B. Cohen, E. Giunchiglia. Can I Have Your Order? Monte-Carlo Tree Search for Slot Filling Ordering in Diffusion Language Models. *ICML* 2026. [\[Link\]](#)
- [3] **M. C. Stoian**, T. Lukasiewicz, E. Giunchiglia. A Survey on Deep Learning Approaches for Tabular Data Generation: Utility, Alignment, Fidelity, Privacy, Diversity, and Beyond. *TMLR* 2026. [\[Link\]](#)
- [4] **M. C. Stoian**, E. Giunchiglia. Beyond the Convexity Assumption: Realistic Tabular Data Generation under Quantifier-Free Real Linear Constraints. *ICLR* 2025. [\[Link\]](#)
- [5] **M. C. Stoian**, S. Dymishi, M. Cordy, T. Lukasiewicz, E. Giunchiglia. How Realistic Is Your Synthetic Data? Constraining Deep Generative Models for Tabular Data. *ICLR* 2024. [\[Link\]](#)
- [6] **M. C. Stoian**. Deep Learning with Requirements in the Real World. *IJCAI* Doctoral Consortium 2024. [\[Link\]](#)
- [7] **M. Stoian**, A. Tatomir, T. Lukasiewicz, E. Giunchiglia. PiShield: A PyTorch Package for Learning with Requirements. *IJCAI* 2024. [\[Link\]](#)
- [8] E. Giunchiglia, **M. C. Stoian**, S. Khan, F. Cuzzolin, T. Lukasiewicz. ROAD-R: The Autonomous Driving Dataset with Logical Requirements. *Machine Learning* 112, 2023. [\[Link\]](#)

Work Experience

Imperial College London

Research Associate

London, UK

2025–present

- **PI:** Dr. Eleonora Giunchiglia
- **Constrained theorem proving:** developing methods that enforce Lean 4 syntax constraints during LLM proof generation and fine-tuning to increase the proportion of syntactically valid proof candidates and reduce wasted inference compute.
- Funded by **Renaissance Philanthropy**.

FiveAI

Research Intern — Computer Vision & Autonomous Vehicles

Oxford, UK

2020–2021

- **Supervisor:** Dr. Tommaso Cavallari
- **3D symmetry detection:** designed a method that estimates planar reflective symmetries on 3D inputs by slicing along the height dimension and feeding slices to a 2D convolutional recurrent regressor, avoiding costly 3D convolutions.

The University of Edinburgh
Research Assistant — NLP & Machine Translation

Edinburgh, UK
2019

- **Supervisor:** Prof. Sharon Goldwater
- **Speech-to-text translation:** showed that ASR pretraining with target-language data augmentation improves translation performance, and that pretrained-ASR word error rate directly predicts translation quality.
- Published at ICASSP 2020.

ETH Zurich
Summer Research Fellow — Network Verification & Program Synthesis

Zurich, Switzerland
2018

- **Supervisors:** Prof. Martin Vechev, Asst. Prof. Dana Drachler Cohen
- **Program synthesis for P4:** built a Python parser for P4 (Programming Protocol-Independent Packet Processors) and used the Z3 solver to encode and synthesise intended packet-processing behaviour.

Software & Patents

Software: PiShield — a PyTorch framework for knowledge-aided ML [github.com/mihaela-stoian/PiShield]

Patent: *3D Perception*, US20240212189A1 (Five AI Ltd, 2024; pending)

Awards

IX-WAI Early Career Grants , Imperial College London	2026
Oxford PhD Runner-up Prize , G-Research	2025
EPSRC Scholarship for Doctoral Studies , University of Oxford	2021–2025
Women in Quant Finance Grant , G-Research	2025
IJCAI Grant , IJCAI-AIJ	2024
Conference Travel Grants , St Hilda's College, Oxford	2023, 2024
Best Paper Award , AI4AD Workshop @ IJCAI	2022

Selected Talks

OxBridge Women in CS Conference , University of Oxford (<i>selected for oral presentation</i>)	2026
Civil Service Leadership Group , Imperial College London (<i>invitation-only</i>) <i>One of only four projects selected; presented to senior UK civil servants, incl. the Head of the Civil Service.</i>	2025
Dagstuhl Seminar on Logic and Neural Networks (<i>invitation-only</i>)	2025
IJCAI Doctoral Consortium	2024

Technical Skills

Languages — Python (*proficient*); C++, Java, Bash/Shell (*familiar*)

Deep learning — PyTorch, PyTorch Lightning, TensorFlow; Hugging Face, verl

ML & data — scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Seaborn; Jupyter

Tooling — Linux/command line, Git, Docker, Slurm, Conda, Weights & Biases, TensorBoard, $\LaTeX$

Mentoring & Teaching

Tutorial: Beyond Soft Penalties: Hard Constraints for Neural Networks with Continuous Outputs — UAI 2026 (*upcoming*)

Co-supervision: A. Godun (MSc, TU Wien); S. Lee (MSc, Imperial); L. Pejic (BSc, TU Wien)

Teaching: Class Tutor, University of Edinburgh (2017–2019) — *Discrete Mathematics; Algorithms & Data Structures; Formal & Natural Languages*

Leadership & Service

Organiser: ROAD++ Workshop & Challenge (ECCV 2024, ICCV 2023); ROAD-R Challenge (NeurIPS 2023)

Reviewing: NeurIPS, ICLR, ICML, IJCAI, NeSy; *Machine Learning* journal

Languages

English (*proficient*) • Romanian (*native*) • German (*elementary*)